

Global Fertility & Energy Consumption

Are we running out of people — or just substituting them with energy?

206 countries · 1990–2023 · World Bank API + OWID Energy Data

The Data: Two Sources, One Story

206 countries · 34 years · 19 variables — built from two public APIs via Python

World Bank Indicators API

9,044

rows pulled

217

countries

8

indicators

Variables selected:

- ▶ Fertility rate (TFR) — primary outcome
- ▶ Adolescent fertility rate
- ▶ Life expectancy at birth
- ▶ Infant mortality rate
- ▶ Total population
- ▶ Urban population %
- ▶ GDP per capita (USD)
- ▶ Female secondary enrollment %

Our World in Data — Energy Dataset

21,791

rows raw

130

raw columns

11

selected

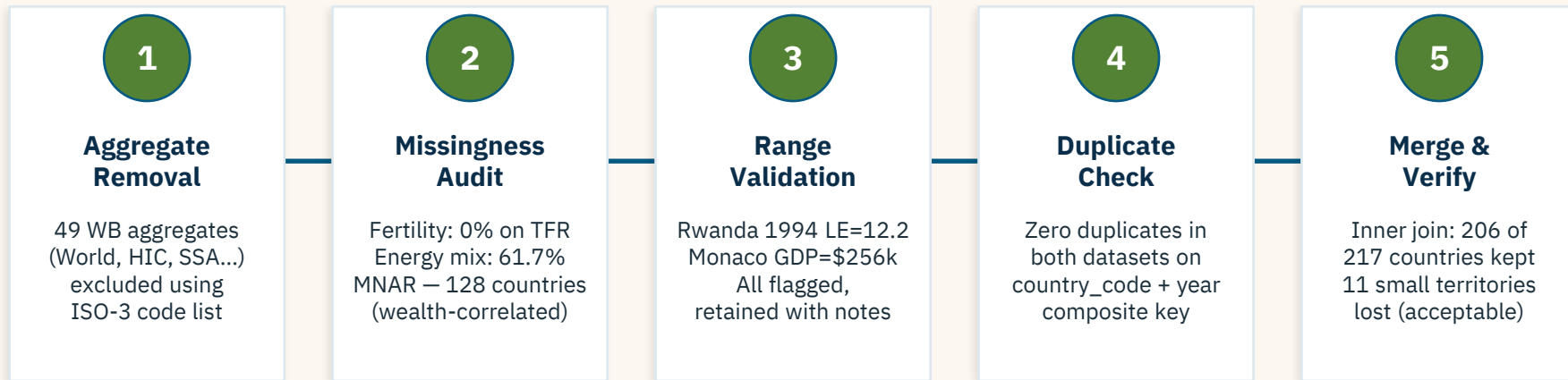
Variables selected:

- ▶ Energy per capita (kWh) — core hypothesis
- ▶ Electricity per capita (kWh)
- ▶ Energy per GDP (intensity)
- ▶ Primary energy consumption (TWh)
- ▶ Fossil share of energy (%)
- ▶ Renewables share of energy (%)
- ▶ Low-carbon share of energy (%)
- ▶ Renewables share of electricity (%)
- ▶ Greenhouse gas emissions
- ▶ GDP (for cross-check)
- ▶ Electricity demand (TWh)

Merged on country_code + year (inner join) → 6,931 rows × 27 columns · 206 countries · 1990–2023

Cleaning, Merging & Quality Checks

Systematic audit before any analysis — 4 check types, 2 datasets



Key missingness findings — what we kept and why

Variable	Missing %	Pattern	Decision
TFR / Life expectancy / Population	0.0%	None	Used in all analysis
Infant mortality	9.7%	Structural (conflict states)	Used; noted in caveats
Female secondary enrollment	33–65%	MNAR — poorer countries	Excluded from models
Energy mix (fossil / renewables share)	61.7%	MNAR — 128 wealthier countries	Analysed separately (n=78)

EDA: Fertility Patterns

Four analytical lenses — distribution, ranking, trend, and correlation

Histograms & Distributions

Section 4.1

Bimodal distribution in 2022: one cluster at TFR 1.0–1.5 (East Asia/Europe), another at 4.0–6.5 (Sub-Saharan Africa)

Top/Bottom Rankings

Section 4.2

Niger (6.8) vs Macao (0.68) — a 10× gap. 13 of 15 highest-fertility countries are in Sub-Saharan Africa

Time Series Trends

Section 4.3

Global median fell from 3.44 (1990) to 1.98 (2023). Two structural breaks dominate: China 1991 and India 2020

Below-Replacement Analysis

Section 4.4

54% of countries and 67% of world population now in below-replacement countries. Crossed ~2019. Population-weighted trend driven by India and China

Correlation analysis (Section 4.5): Spearman ρ — infant mortality +0.81 · life expectancy -0.78 · GDP -0.77 · all computed on 2022 snapshot (n=206)

EDA: Energy Analysis & SQL Queries

Energy EDA confirmed the hypothesis visually — SQL formalised it analytically

Energy EDA — Sections 4.8–4.13

Distributions

Energy per capita right-skewed (max 195k kWh — Qatar). Log transformation applied in Section 5.5

Scatter plots

Energy vs fertility: strong negative relationship ($\rho = -0.73$).
Renewables share: flat ($\rho = 0.07$)

Time trends

Global median energy +55% (1990→2023) while fertility fell −42% — opposite directions across all 34 years

Group comparison

Below-replacement countries: 28,264 kWh median vs 3,111 kWh above — a 9× energy gap confirmed visually

Extended correlation

Heatmap with all 19 variables: energy cluster independent of renewables type — only quantity matters

10 SQL Queries — SQLite In-Memory DB

Q1–2 GROUP BY

Fertility trend by year; energy coverage by year

Q3 INNER JOIN

Core merged dataset: fertility + energy on country+year

Q4 LEFT JOIN

Anti-join: 11 countries in WB but not OWID

Q5 INNER JOIN

High-energy, above-replacement outliers (18 countries)

Q6 RANK()

Window: rank countries by fertility within each year

Q7 LAG()

Window: year-over-year energy change per country

Q8 Subquery

Above-average energy AND below-replacement (58 countries)

Q9 Self-join + Subquery

Fastest energy growth + fertility decline 1990–2022

Q10 GROUP BY + CASE WHEN

Income gradient: 49× energy gap, monotonic across groups

The Global Fertility Transition

Median country fertility crossed below replacement (2.1) around 2019

54%

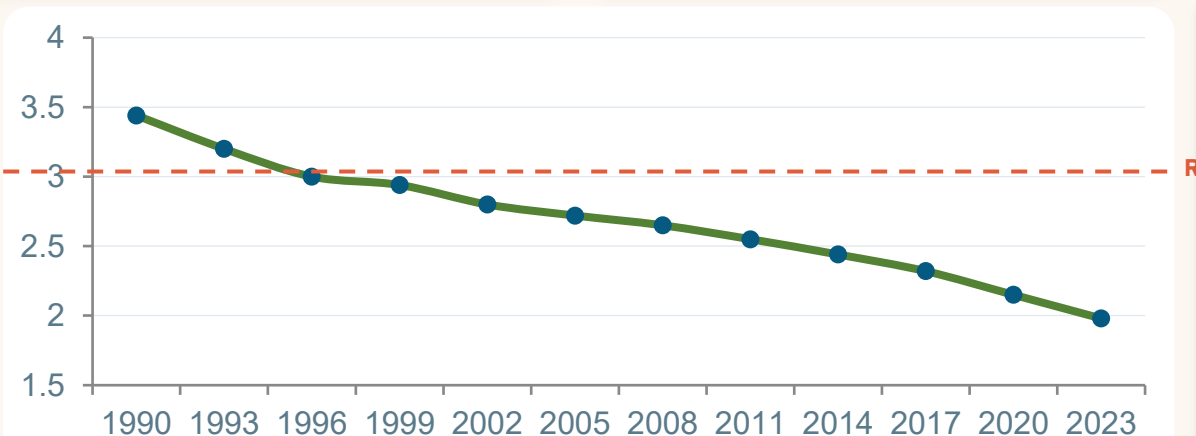
of countries below
replacement in 2022

67%

of world population
in below-replacement countries

3.44 → 1.98

global median TFR
1990 to 2023



Two structural breaks

1991:

China crosses below 2.1
+1.16B people

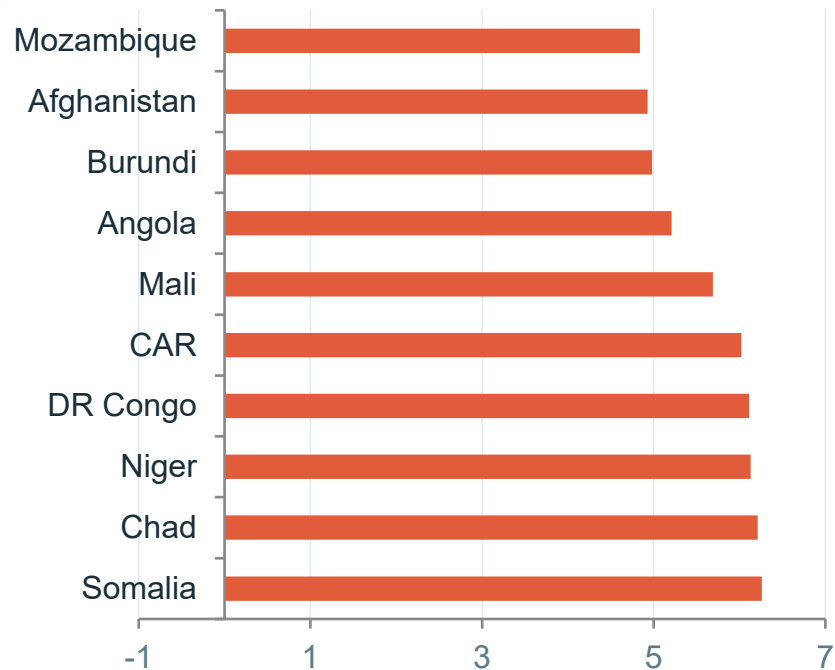
2020:

India crosses below 2.1
+1.53B people

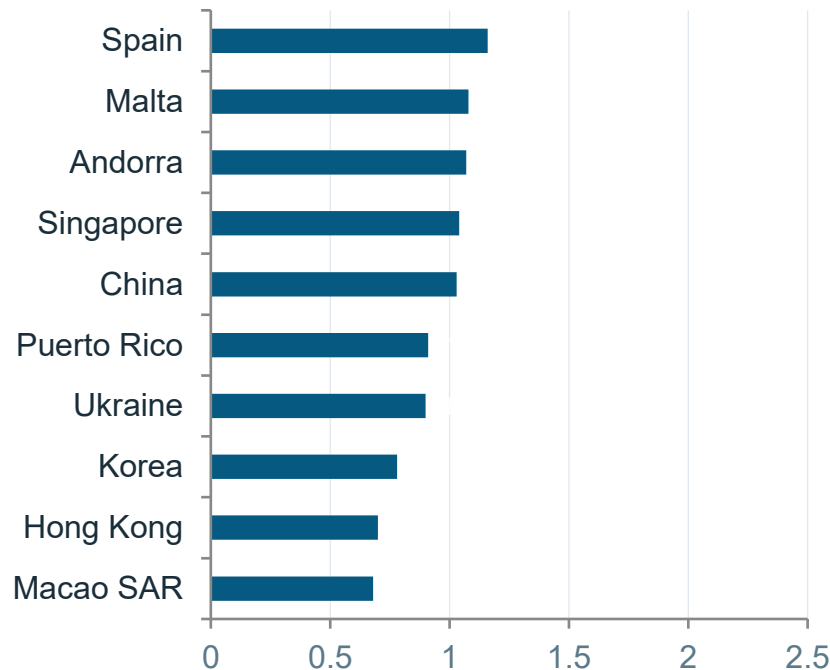
A Tale of Two Worlds

Fertility extremes are geographically concentrated — and moving in opposite directions

Highest TFR — 2022



Lowest TFR — 2022

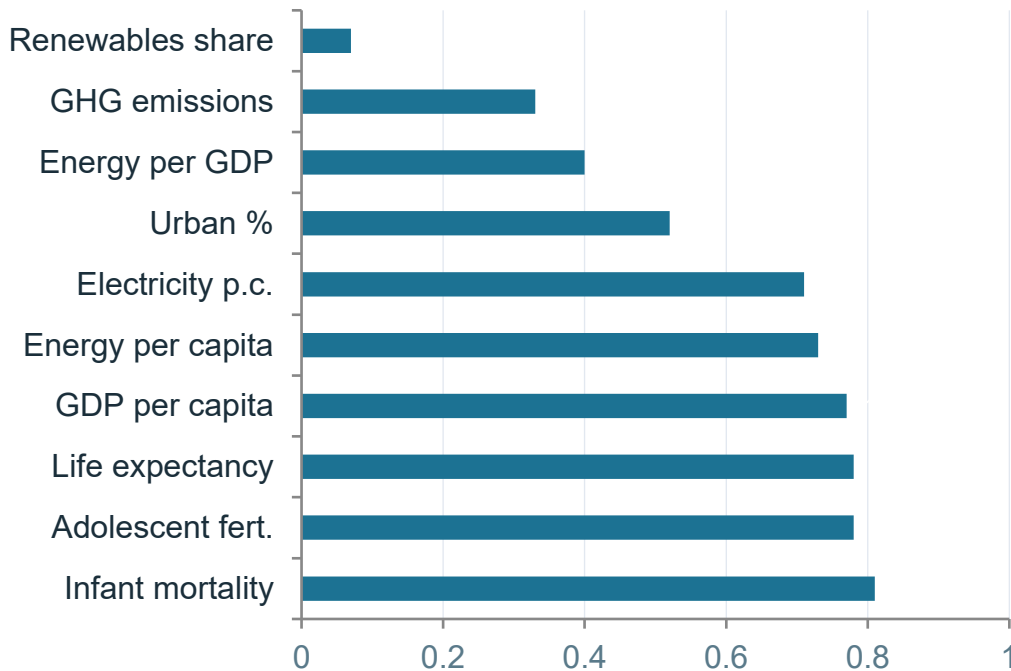


Sub-Saharan Africa dominates the high end (13/15 countries). East Asia dominates the low end — Korea at 0.78 is below any level in recorded demographic history.

Why Fertility Falls: The Development Ladder

Demographic Transition Theory confirmed — every indicator moves in lockstep

Spearman ρ with fertility rate (2022)

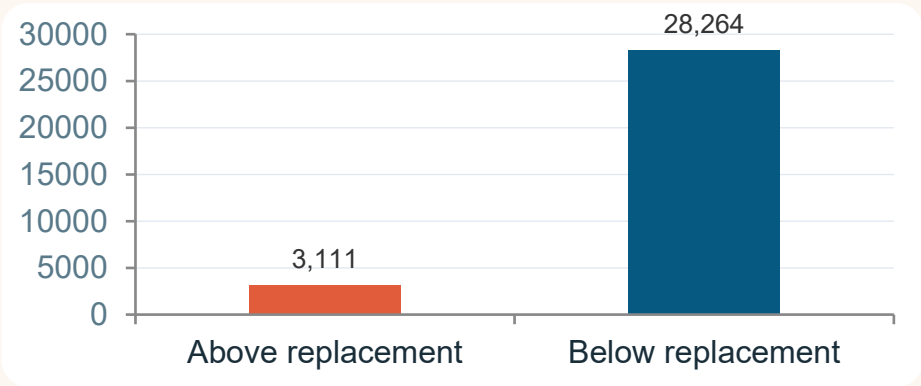


- ① Child survival is the strongest predictor — parents have fewer children when babies survive ($\rho = 0.81$)
- ② Energy per capita rivals GDP ($\rho = -0.73$ vs -0.77) — suggesting independent energy effect
- ③ Renewables share = zero correlation ($\rho = 0.07$) — type of energy is irrelevant, only quantity matters
- ④ All development indicators cluster together — multicollinearity is a feature, not a bug

The Energy–Fertility Hypothesis

As societies substitute energy for human labour, fertility declines — confirmed cross-sectionally and longitudinally

Median energy use by fertility group — 2022



Below-replacement countries use 9× more energy per person

Energy-fertility gradient across income groups

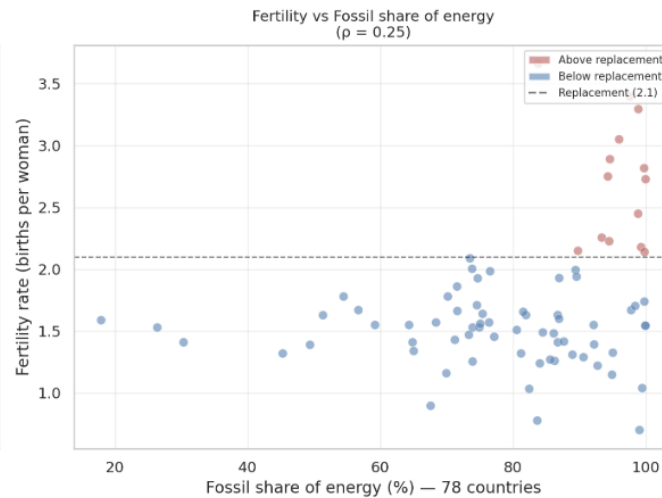
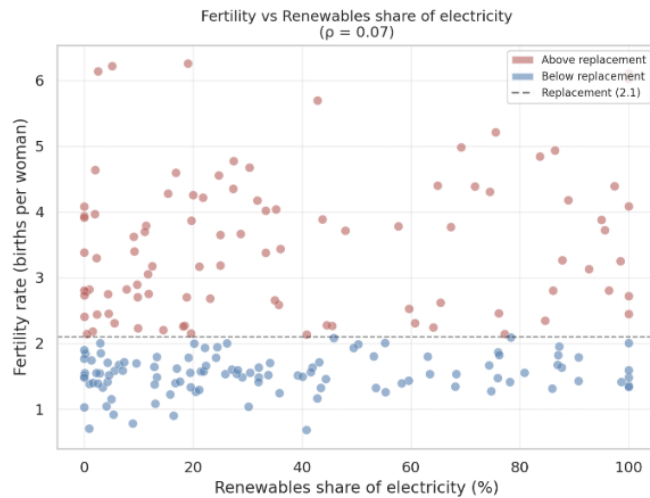
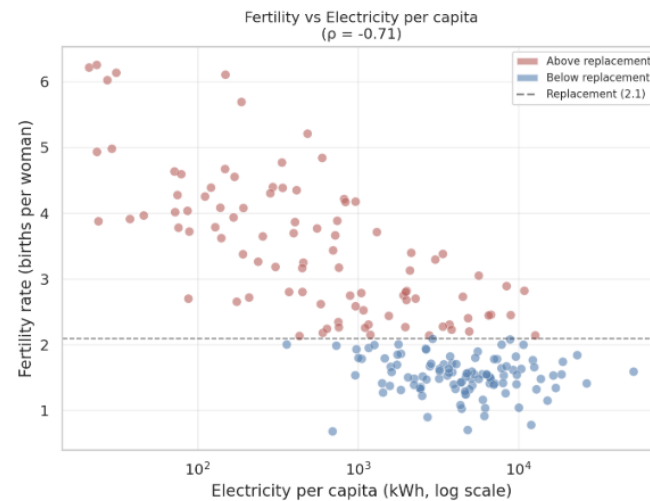
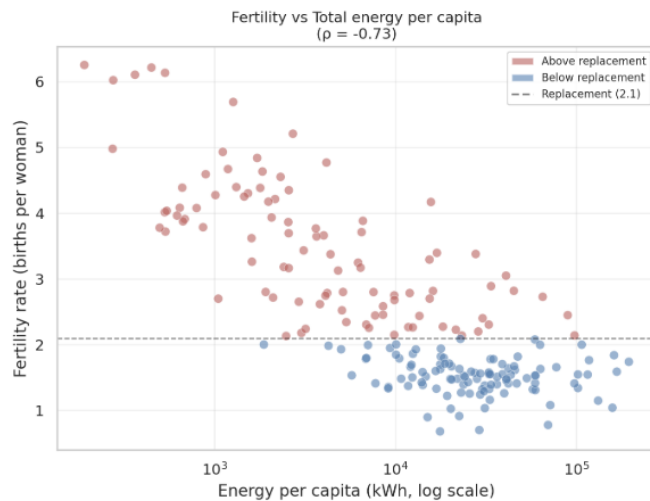
Longitudinal evidence (SQL Query 9)

25 countries with above-average energy growth
ALL showed significant fertility decline 1990–2022

Bhutan	+1,970% energy	−4.12 TFR
Iran	+181% energy	−3.23 TFR
India	+165% energy	−2.05 TFR

Income group	Avg fertility	Avg energy/capita	Avg life exp.
Low income (<\$1,500)	4.34	1,291 kWh	62.2 yrs
Lower-middle (\$1.5k–\$7k)	2.72	9,598 kWh	70.4 yrs
Upper-middle (\$7k–\$25k)	1.83	24,768 kWh	75.0 yrs
High income (>\$25,000)	1.52	63,186 kWh	80.5 yrs

Fertility rate vs energy indicators, 2022



What the Models Confirm

Two regression models — same story

Linear Regression

$R^2 = 0.71$

RMSE = 0.73 births/woman
183 countries

Logistic Regression

AUC = 0.925

Accuracy = 85%
40-country test set

Energy per capita coefficient = -0.43 (consistent across all model specs)

- 1 More than half the world is now below replacement — driven by China (1991) and India (2020)
- 2 Energy quantity predicts fertility as strongly as GDP — an independent effect beyond income
- 3 Energy type is irrelevant — only access and quantity matter (renewables $p = 0.07$)